PROJECT AND TEAM INFORMATION

Project Title

Smart Traffic Management System Using OS-Based Real-Time Process Scheduling

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Traffic congestion is one of the problems that affect most cities. Busy intersections experience the problem most when the number of vehicles varies over different days.
Old-fashioned traffic signals usually stick to constant schedules. It might result in unnecessary waiting when one road is mostly empty while the other is really congested.
Besides causing inconvenience by making travel time longer, the traffic congestion leads to more fuel consumption and increased vehicle emissions. The literature review shows that the use of traffic signals which can be adjusted to the actual traffic conditions can bring traffic flow improvement and the elimination of unnecessary waiting time.
One of the selected articles uses SUMO and Deep Reinforcement Learning (DRL) to adapt traffic-signal cycles by taking into account waiting time and CO emission.
The objectives of the project were to create a simulation traffic signal control system of an intersection and then analyze the effect of different types of traffic schedule on the performance of traffic. The Sumo will be used as the main program for running simulations, Python will be means for interfacing of the simulator with the control logic. An AI agent will be the main component of our decision making.
Though, the concepts related to Operating System will be used on the control side namely the priority scheduling, synchronization, and fairness. The traffic data, signal decisions, and results of experiments are all going to be stored in a DBMS.
Main motivation is combining different concepts in one project and finding out if an adaptive system really can outperform the fixed-time one in traffic flow management.

State of the Art / Current solution

Present day traffic-signal equipment rely mostly on fixed-time signals or rule-based control, where signal timings of the traffic lights are decided at prior.
Such control systems are straightforward, yet might be ineffective in situations where there is a sudden high traffic density.
A lot of recent work has gone into research of traffic control which is adaptive to the actual situation, using traffic sensors, machine learning and reinforcement learning.
Gupta et al. (2024) used SUMO to simulate a four-way intersection and proposed a Deep Reinforcement Learning (DRL)-based approach for adaptive traffic signal control. Their focus was to reduce vehicle waiting time while also considering CO emissions.
Kumaran et al. (2025) studied traffic-density detection and adaptive signal optimization using YOLOv5, reinforcement learning and other optimization techniques. The study evaluated the approach using things like waiting time, vehicle clearance rate and traffic-flow rate.
For our project, we are going to use these methods, make them fit with our educational environment and implement a system that integrates traffic simulation, AI-based control, OS scheduling and DBMS-based analysis.

Project Goals and Milestones

The motivation of this project is to develop a Virtual Traffic Signaling Management system, capable of monitoring the varying traffic situations, and control the operation of signals and at the same time illustrate some ideas of Operating System and DBMS.

Goals:

1.Design a four-way virtual intersection and traffic situations with SUMO.
2.Integrate SUMO with Python to traffic data retrieval and signal-control command sending to the simulator.
3.Implement a control layer based on the Operating System concepts like priority scheduling, queue management, synchronization and aging to prevent starvation.
4.Devise and test an AI-based reinforcement-learning agent, starting out with DQN as the approach of choosing a policy from a deep neural network, for changing traffic control as the circumstances.DQN/DRL-based traffic signal control is described and compared in the papers that are cited here.
5.Develop a relational databases which is able to store such information as vehicle data, traffic conditions, signal state changes, decisions made by AI, and results of the study.
6.Determine the difference between the methods such as fixed-time, OS-based and AI-assisted traffic control.
7.Evaluate the performance of traffic using these indices: average waiting time, queue length throughput and, if the simulation is able to do it, CO emissions.

Initial milestones:

Milestone 1: Read selected research papers and decide the final system design.
Milestone 2: create SUMO road network, intersection and traffic scenarios.
Milestone 3: Enable communication between Python and SUMO.
Milestone 4: complete OS scheduling and synchronization.
Milestone 5: Put together and run/train/test the AI agent.
Milestone 6: Build a DBMS and perform SQL analysis.
Milestone 7: Conduct experiments compare the results and complete the final report and presentaion

Project Approach

The system will create SUMO (Simulation of Urban MObility) as the virtual environment. Besides that, different traffic scenarios will be designed with SUMO, like no traffic condition, one-way heavy traffic and more balanced traffic conditions. The system will create different scenes of traffic in the road network, intersection traffic light, vehicle routes etc,A Python script will be the main module acting as the bridge between SUMO and the rest of the modules.

It will receive from the simulation inputs such as vehicle counts, queue lengths and waiting information. Then, these inputs will be used for decision-making process by the module.

The OS module traffic-control requests will be modelled as tasks that are schedulable. Priority scheduling, synchronization aging are mechanisms employed to ensure traffic-control tasks are handled safely. For instance, emergency requests would get higher priority than long-awaited traffic.

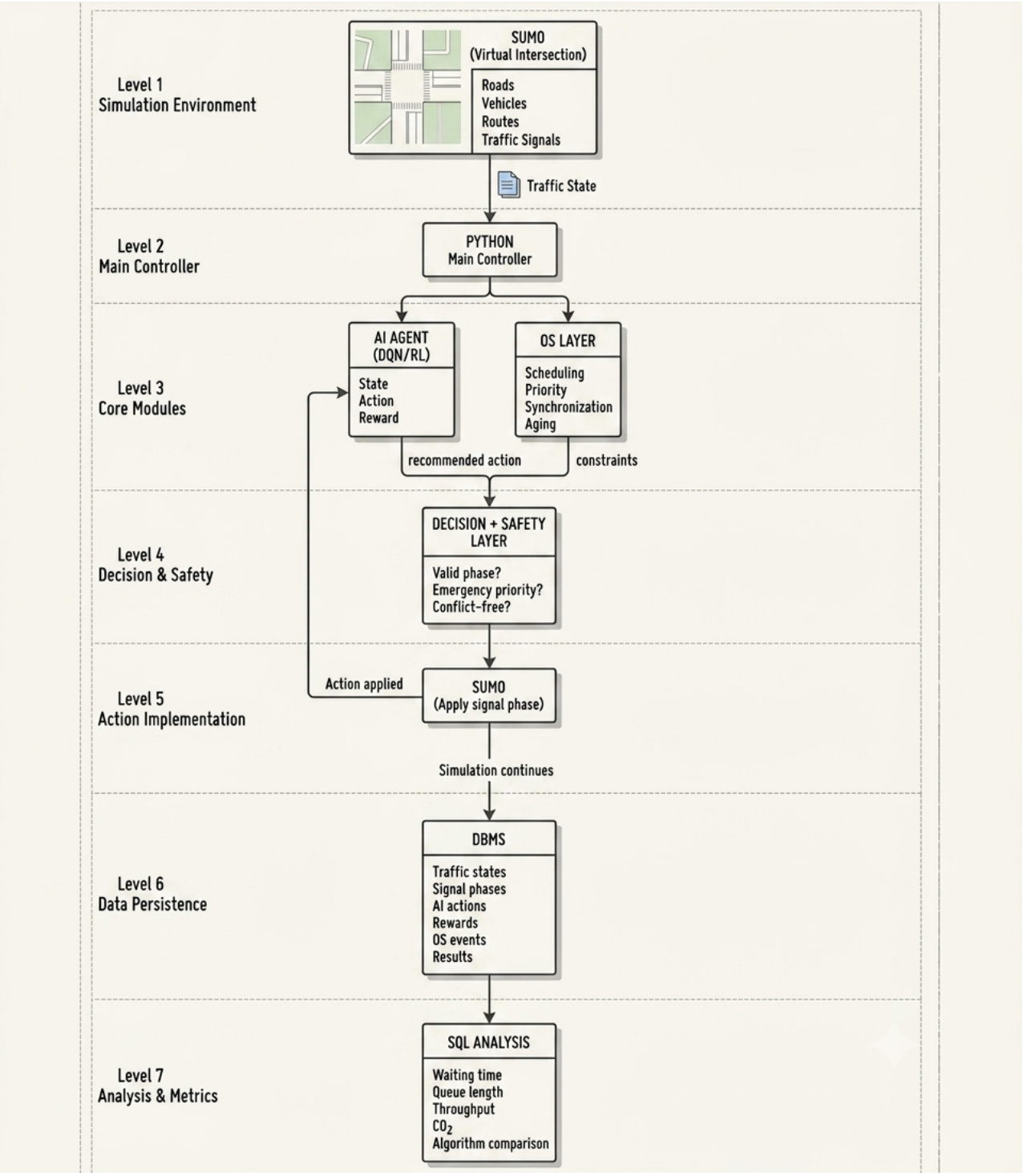
Reinforcement learning-based strategy will be adopted by the AI part, at first, it will implement DQN. At first the agent will only be able to observe the traffic status and, as it learns, the signal actions that have been defined.

Once the actions have been selected, they will become a part of the simulation in SUMO and so the traffic consequences will be a feedback for training the model.

The DBMS will serve mainly two needs; storing of traffic states, signal changes, vehicle details, AI decisions and experiment results first, and on the other it enables analysis of these data through SQL queries such as computing averages or comparison of the performance of the diverse control methods.

It is planned that SUMO, Python, TraCI, DQN/RL, MySQL and data-visualization tools are the key technologies to implement the system.

System Architecture (High Level Diagram)



Project Outcome / Deliverables

At the end of this project, the team will have built a functioning virtual traffic signal management system that can simulate changing traffic conditions and try out different signal-controller schemes, Our main work products will be
1.A SUMO intersection set-up which will run cars on the streets and change signals.
2.A Python program for SUMO connection and communication to the rest of the system.
3.An Operation system layer to implement priority scheduling, synchronization and fairness principles.
4.An AI traffic signal controller that adapts to traffic flow based on reinforcement learning/DQN.
5.A relational DBMS storing all the simulation data: vehicles traffic signals... and the actual experiment data.
6.SQL queries and performance visuals.
7.An experiment study comparing different methods of control: fixed-time, OS-based, and AI-assisted.
8.The results of the experiments will be based on factors like waiting time, queue length, and throughput, and the CO emissions, if the simulator will include them.

Assumptions

The simulation-based development approach will be used in this project instead of a real traffic-control. implementation. Only a small number of traffic-light junctions and vehicle models will be involved. to make the system easy to handle. In the first place, SUMO will be used to create the flow of traffic using pre-programmed scenarios of different densities.
The traffic lights cycles will be set ahead so that the human-free machine can only choose, those actions that are safe and valid. The team will also work on getting traffic-flow performance data from SUMO. Carbon monoxide emissions will be taken into account only if there are emission-level measures suitable for that in the SUMO setup that is finally chosen.

References

[1] A. Gupta, A. Agarwal, and V. A. Bohara, "Adaptive Traffic Signal Cycle Control for Green City Traffic Management," TENCON 2024 – 2024 IEEE Region 10 Conference, 2024. DOI: 10.1109/TENCON61640.2024.10903119.
[2] K. Kumaran, K. K. S. Kindinti, S. A. Supriya, and H. S. V. T., "Traffic Density Detection And Signal Optimization Using YOLOv5 And ANT Colony Optimization," 2025 International Conference on Data Science, Agents, and Artificial Intelligence (ICDSAAI 2025), 2025. DOI: 10.1109/ICDSAAI65575.2025.11011701.